

providing intuitive libraries and cross-platform capabilities, these tools support the development of scalable and reproducible quantum machine learning solutions.

Quantum neural networks (QNNs) and quantum kernel methods

One of the most promising areas in quantum machine learning is the development of quantum neural networks (QNNs). QNNs leverage quantum mechanics to process information in ways that classical neural networks cannot, potentially leading to breakthroughs in fields like image recognition, natural language processing, and other data-intensive applications. QNNs are still in the early stages of development, but they offer potential advantages such as increased parallelism and faster training times for certain problem types. For example, a QNN can be implemented in Qiskit or PennyLane using specialised quantum

layers that encode data into quantum states, allowing models to learn complex relationships in data more efficiently.

Another essential tool in quantum-enhanced machine learning is the quantum kernel. Quantum kernels are particularly useful for support vector machines (SVMs) and other kernel-based learning methods that rely on similarity measures. Quantum SVMs can process information in high-dimensional spaces and are expected to outperform classical counterparts in tasks involving large datasets with intricate patterns. By utilising quantum kernels, machine learning models can better identify non-linear relationships in data. Tools like Qiskit's machine learning module offer functionality for building and training quantum kernel-based models, making it accessible for real-world applications such as anomaly detection and data classification.

Figure 2 compares classical and quantum neural network architectures.

Hybrid quantum-classical algorithms

Since quantum computers are still limited in terms of scalability and error rates, hybrid quantum-classical algorithms have emerged as a practical solution. Hybrid algorithms use quantum circuits for parts of the computation, where quantum computing offers advantages (e.g., specific optimisation tasks), while classical systems handle the rest.

Two popular hybrid algorithms are the Variational Quantum Eigensolver (VQE) and the Quantum Approximate Optimization Algorithm (QAOA).

VQE: VQE is commonly used for solving optimisation problems, especially those related to chemistry and materials science. By adjusting parameters within a quantum circuit, VQE minimises the energy of a system, which can be applied to find optimal solutions in complex machine learning tasks.

QAOA: QAOA is another hybrid algorithm suited for combinatorial

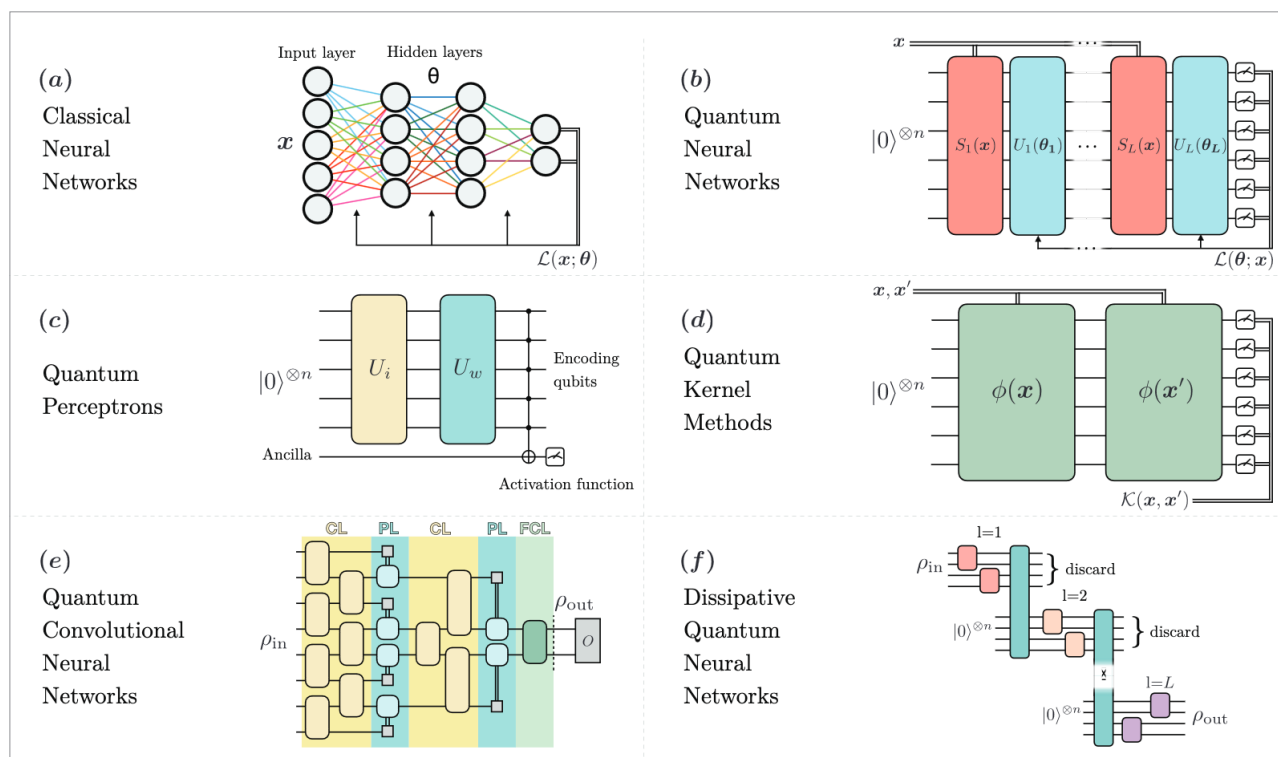


Figure 2: Comparison of classical and quantum neural network architectures